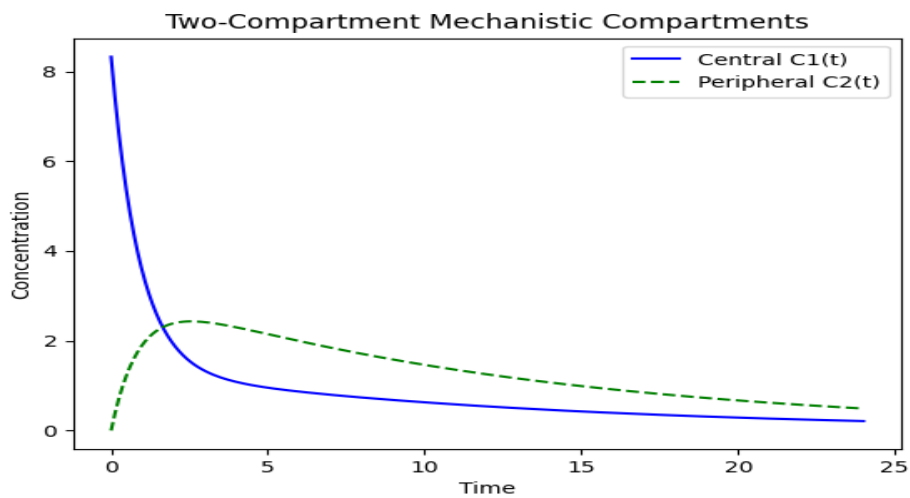


Note: Auto-demoted to two-compartment for Subject 1 ( $\Delta AICc < 2$  and/or poor 3c rate separation / weak tail or  $n < 8$ ).

# Subject 1

## Central vs. Peripheral Compartments



## Study Metadata

| Field        | Value        |
|--------------|--------------|
| study_id     | UserStudy1   |
| species      | Mus musculus |
| route        | IV bolus     |
| dose         | 100          |
| model        | three        |
| time_units   | h            |
| conc_units   | a.u.         |
| allow_demote | True         |
| k10          | 0.35         |
| k12          | 0.6          |
| k21          | 0.25         |
| k13          | 0.35         |
| k31          | 0.12         |
| V1           | 0.04         |

|         |      |
|---------|------|
| V2      | 0.06 |
| V3      | 0.1  |
| subject | 1    |

## Data Summary

|                                |                |
|--------------------------------|----------------|
| n                              | 42             |
| Time span (min → max)          | 0.00 → 24.00 h |
| C_max (observed)               | 8.31 a.u.      |
| Global t½ (log-linear)         | 3.10 h         |
| m_early / m_late  (log-slopes) | 1.90           |

## Fit Results (with 95% CI)

| Parameter | Estimate | 95% CI         |
|-----------|----------|----------------|
| A         | 6.29     | [6.11, 6.46]   |
| alpha     | 1.03     | [0.989, 1.07]  |
| B         | 2.04     | [1.87, 2.21]   |
| beta      | 0.201    | [0.189, 0.214] |
| Vd_ss     | 0.1      |                |
| Cl        | 0.014    |                |

## Goodness-of-Fit

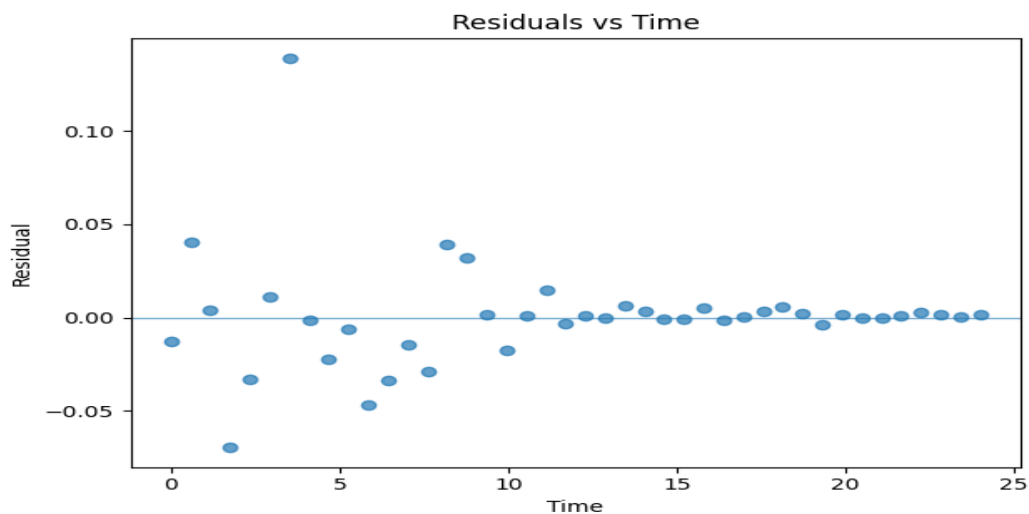
|                |        |
|----------------|--------|
| R <sup>2</sup> | 1.000  |
| AIC            | -289.4 |
| AICc           | -288.3 |

## Model Selection Diagnostics

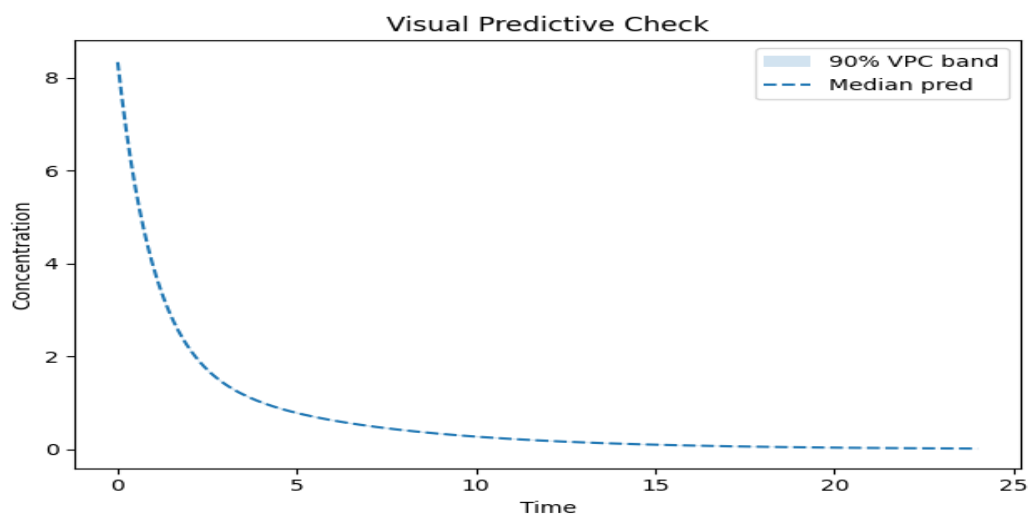
| Criterion       | AICc (linear) |
|-----------------|---------------|
| AICc (2c)       | -288.29       |
| AICc (3c)       | -277.43       |
| ΔAICc (3c – 2c) | 10.86         |

|                   |       |
|-------------------|-------|
| rate_sep_rel (3c) | 0.417 |
|-------------------|-------|

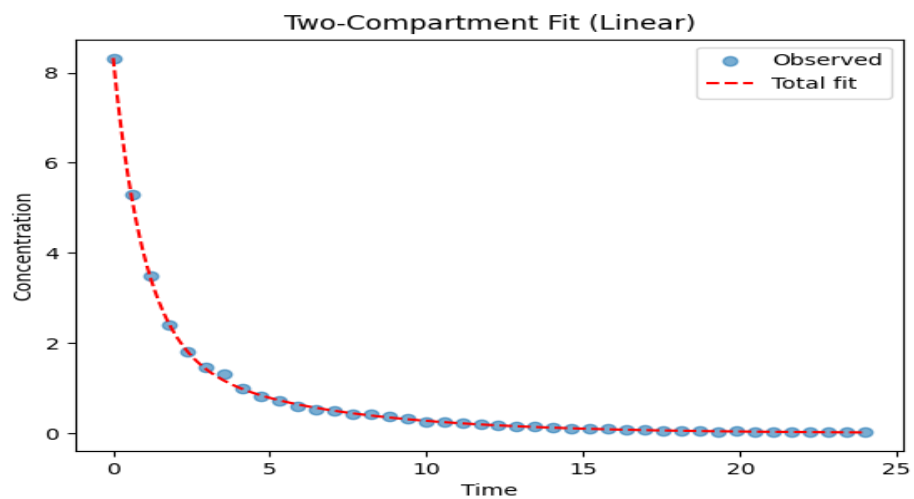
## Residual Diagnostics



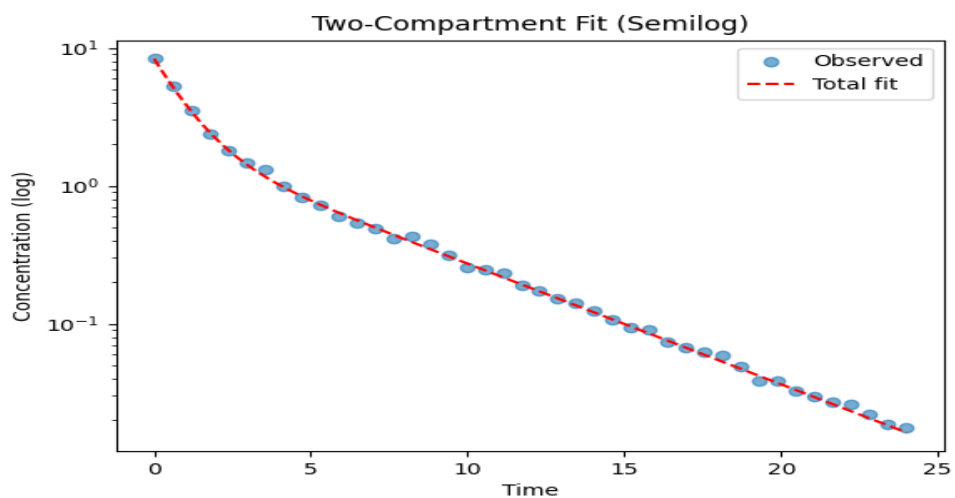
## Visual Predictive Check (5–95%)



## Linear Fit



## Semilog Fit



# Dosing Timeline

